A direct proof of the complete Crouzeix inequality

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Abstract

Every square matrix A comes with a compact convex subset $W(A)$ of the complex plane, called its numerical range. The complete Crouzeix conjecture says: if you plug A into a matrix-valued polynomial P, the result is never larger than twice the biggest value of P on $W(A)$, and this is uniform in all matrix dimensions. We give a self-contained direct proof. It uses an exterior conformal map of an analytic convex domain containing $W(A)$, a positive kernel on the unit circle, and a two-by-two positive block comparison. Degenerate cases, where $W(A)$ is a single point or a line segment, are included.

1 Introduction

How to read this paper

The first section states the main theorem and explains what is going on. Sections **??**–**??** prove it. The Appendix collects three background items: the exterior conformal map, a short recall of Schwarz reflection, and the classical Jordan-domain theorem references.

If you are a strong high-school student, you already know enough linear algebra and complex numbers to read the Introduction and to see the main ideas in Section **??**. The one technical part is the conformal map in Section **??** and Appendix **??**. You may treat Appendix **??** as a black box on a first read.

The numerical range

Fix a positive integer n and a matrix $A \in M_n(\mathbb{C})$. A vector $u \in \mathbb{C}^n$ with $u^*u = 1$ is a *unit vector*. Its *Rayleigh quotient* is the complex number u^*Au. The *numerical range* of A is the set

$$W(A) \;=\; \{\, u^*Au \,:\, u \in \mathbb{C}^n,\; u^*u = 1 \,\}. \tag{1.1}$$

The numerical range of A is always a compact convex subset of $\mathbb{C}$, and it always contains the spectrum of A: a unit eigenvector u of an eigenvalue λ satisfies $u^*Au = \lambda u^*u = \lambda$.

Two extreme examples are useful.

- If $A = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ is diagonal with real entries, $W(A)$ is the interval $[\min \lambda_i,\; \max \lambda_i]$ on the real axis.
- If $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, then $|u^*Au| = 2|u_1u_2| \le |u_1|^2 + |u_2|^2 = 1$ for every unit u, with equality whenever $|u_1| = |u_2| = 1/\sqrt{2}$. Convexity then forces $W(A)$ to be the closed unit disk $\overline{\mathbb{D}}$. Notice that the spectrum is just $\{0\}$, yet the numerical range is the whole disk. This is the simplest example where nonnormality shows up in $W(A)$ but not in the spectrum.

For a normal matrix (one with $AA^* = A^*A$), the spectral theorem gives a unitary U with $A = U\Lambda U^*$, and then u^*Au is a convex combination of the eigenvalues. So $W(A)$ is exactly the convex hull of the eigenvalues. For a general matrix, this statement fails badly: the numerical range is genuinely larger than the spectrum, and that extra size is what makes the spectral-set constant 2 possible.

The main theorem

We allow *matrix-valued* polynomials. If $P(z) = \sum_{k=0}^{d} B_k z^k$ with coefficient matrices $B_k \in M_m(\mathbb{C})$, define

$$P[A] \;=\; \sum_{k=0}^{d} A^k \otimes B_k \;\in\; M_n(\mathbb{C}) \otimes M_m(\mathbb{C}) \;\cong\; M_{nm}(\mathbb{C}). \tag{1.2}$$

Here $\otimes$ is the Kronecker product: the factor A^k acts on $\mathbb{C}^n$ and the factor B_k acts on $\mathbb{C}^m$, and the two actions commute. Equivalently, $P[A]$ is obtained by replacing the scalar variable z with the matrix A, keeping the coefficients B_k in the exact order written. The two tensor factors are the *base* $\mathbb{C}^n$ and the *coefficient* $\mathbb{C}^m$.

All norms are the operator norms from the Euclidean inner products and their tensor products (equivalently, the largest singular value). The Hilbert–Schmidt norm of a matrix C is denoted $\|C\|_2 = (\operatorname{tr} C^*C)^{1/2}$.

Theorem 1.1 (Complete Crouzeix inequality)**.** *Let $n, m \geq 1$, let $A \in M_n(\mathbb{C})$, let $d \geq 0$, and let $P(z) = \sum_{k=0}^{d} B_k z^k$ with $B_k \in M_m(\mathbb{C})$. Then*

$$\|P[A]\| \;\leq\; 2 \max_{z \in W(A)} \|P(z)\|. \tag{1.3}$$

The constant 2 is optimal: it cannot be lowered uniformly in n, m, d.

The word *complete* refers to the uniformity in the coefficient dimension m. A scalar proof (the case $m = 1$) does not automatically extend, because scalar coefficients commute but matrix coefficients need not: $F(z)G(z)$ and $G(z)F(z)$ are different $M_m(\mathbb{C})$-valued functions in general, and $F[A]G[A]$ is not $G[A]F[A]$. The proof below keeps these two products separate all the way through, which is why it works uniformly in m.

Why the constant cannot be less than 2

Take $n = 2$, $m = 1$, $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, and $P(z) = z$. The eigenvalue 0 of A has a two-dimensional generalised eigenvector space, and $\|A\| = 2$. Meanwhile the previous subsection showed that $W(A)$ is the closed unit disk, so $\max_{W(A)} |P(z)| = \max_{W(A)} |z| = 1$. The ratio $\|P[A]\| / \max_{W(A)} |P| = 2$, so no universal constant below 2 works. The rest of the paper proves that 2 is enough.

Roadmap of the proof

The argument has five steps.

1. **Enclose.** Place $W(A)$ strictly inside a bounded convex open set Ω whose boundary is a smooth real-analytic Jordan curve. This lets us use conformal mapping. The final section shows how to remove this hypothesis.

2. **Coordinate.** Build an exterior conformal map $h : \{\lambda : |\lambda| > 1\} \to \mathbb{C} \setminus \overline{\Omega}$. On the unit circle $|\lambda| = 1$, h parametrises $\partial\Omega$, so every boundary integral becomes a circle integral.

3. **Kernel.** Expand the Cauchy kernel $q(\lambda)/(h(\lambda) - z)$ at infinity, where $q(\lambda) = \lambda h'(\lambda)$. This produces polynomials $b_k(z)$ (the *Faber polynomials* of the exterior coordinate), and a positive kernel $K(\lambda, \mu)$ on the unit circle. Positivity of K is where convexity of Ω enters the proof.

4. **Coefficient bound.** From K, deduce that for every matrix-valued polynomial $G = \sum b_k C_k$,

$$\sum_k \|C_k\|_2^2 \;\le\; \int_{\mathbb{T}} \|G(h(\mu))\|_2^2 \, d\tau(\mu) \;\le\; \|C_0\|_2^2 + 2 \sum_{k \ge 1} \|C_k\|_2^2. \tag{1.4}$$

   The factor of 2 on the right is the sole source of the constant 2 in Theorem **??**. This step uses the positivity of K in a decisive way.

5. **Functional analysis.** Let $R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}$ be the exterior resolvent. Positivity of $R + R^*$ (a consequence of $W(A) \subset \Omega$) lets a two-by-two positive block comparison pass (**??**) to the single polynomial $F[A]$ and force $\|F[A]\| \le 2$ whenever $\|F\| \le 1$ on $\partial\Omega$. Then shrink Ω down to $W(A)$.

Prerequisites and notation

We use:

- Linear algebra: unitary matrices, tensor products, singular values, operator norm $\|\cdot\|$ and Hilbert–Schmidt norm $\|\cdot\|_2$.
- Complex analysis: holomorphic functions, conformal maps, the Riemann mapping theorem, Schwarz reflection. A short self-contained proof of the exterior collar is in Appendix **??**.
- Fourier analysis on the unit circle $\mathbb{T} = \{|z| = 1\}$, equipped with the normalised measure $d\tau(e^{it}) = dt/(2\pi)$. Every constant function of modulus one has integral one; every function with only positive or only negative Fourier modes has integral zero.

Historical context

Crouzeix asked the scalar question in 2004 and proved a universal complete bound of 11.08 in 2007 [**?**, **?**]. The exterior-kernel method traces back to Delyon and Delyon [**?**]. Crouzeix and Palencia obtained the complete bound $1 + \sqrt{2}$ [**?**], and Ransford and Schwenninger simplified the final step [**?**]. Badea, Crouzeix and Delyon proved the complete constant 2 for all base matrices of order 2 [**?**], and Åhag, Czyż, Virtanen proved the complete constant 2 for base matrices of order 3 [**?**]. The scalar case $m = 1$ was settled recently by Jin [**?**], Lorist and Schwenninger [**?**], and Luo [**?**]. A structural proof of the complete case is given in the companion preprint [**?**]; the exterior kernel, marginal identities and positive block comparison below have counterparts there. The point of the present paper is to give a direct and self-contained argument for the *complete* case, with all matrix dimensions independent.

2 Exterior coordinates and Faber coefficients

Throughout this section, $\Omega \subset \mathbb{C}$ is a bounded open convex set whose boundary is a *regular real-analytic Jordan curve*: a simple closed curve parametrised real-analytically with nowhere-zero derivative. The unit circle is $\mathbb{T} = \{\lambda : |\lambda| = 1\}$. The exterior unit disk is $\mathbb{C} \setminus \overline{\mathbb{D}} = \{\lambda : |\lambda| > 1\}$. Every circle integral uses $d\tau(e^{it}) = dt/(2\pi)$, so constant functions of modulus one have integral one.

The exterior coordinate

By Lemma **??** in the Appendix, there is a number $0 < r_0 < 1$ and a map

$$h(\lambda) \;=\; c\lambda + c_0 + \sum_{j\geq 1} c_j \lambda^{-j}, \qquad c \neq 0, \tag{2.1}$$

meromorphic on $\{\lambda : |\lambda| > r_0\} \cup \{\infty\}$, one-to-one there as a map of the Riemann sphere, with a simple pole at infinity, mapping $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$ and $|\lambda| = 1$ onto $\partial\Omega$. This is the *exterior conformal map*, or *exterior coordinate* of Ω.

The Appendix gives a complete derivation from Riemann's mapping theorem. The reader who wants to accept it can take (**??**) and the statements below as basic inputs.

The supporting line and the outward normal

Define

$$q(\lambda) \;=\; \lambda h'(\lambda). \tag{2.2}$$

The two partial derivatives of h at a point $e^{i\theta} \in \mathbb{T}$ are $\partial_\theta h = iq$ (tangent to the boundary) and $\partial_r h = q$ where $r = |\lambda|$ (radial direction, pointing outward from $\overline{\Omega}$). So $q(\lambda)$ is a nonzero outward normal vector at $h(\lambda) \in \partial\Omega$, and $iq(\lambda)$ is a positively oriented tangent vector.

Because Ω is convex, the entire set $\overline{\Omega}$ lies on the inner side of the supporting line through $h(\lambda)$ with normal direction $q(\lambda)$:

$$\Re\Big(\overline{q(\lambda)}\big(h(\lambda) - z\big)\Big) \;\geq\; 0 \qquad (\lambda \in \mathbb{T},\ z \in \overline{\Omega}). \tag{2.3}$$

This is the only geometric input from convexity, and it is what makes the kernel below nonnegative. Figure **??** shows the picture.

$$\begin{array}{ccc} \mathbb{T} \text{ exterior} & \xrightarrow{\;h\;} & \text{exterior of } \overline{\Omega} \\ \lambda & & h(\lambda),\ \text{outward normal } q(\lambda) \end{array}$$

Figure 1: Schematic. The exterior conformal map h sends the exterior unit circle onto the exterior of $\overline{\Omega}$, one-to-one and onto. At $\lambda \in \mathbb{T}$, the vector $q(\lambda) = \lambda h'(\lambda)$ points outward from $\overline{\Omega}$. Convexity of Ω forces every $z \in \overline{\Omega}$ to lie on the inner side of the supporting line $\Re(\overline{q(\lambda)}(w - h(\lambda))) = 0$.

Faber polynomials

Define scalar polynomials $b_k(z)$, $k \geq 0$, by the expansion at infinity

$$\frac{q(\lambda)}{h(\lambda) - z} \;=\; \sum_{k\geq 0} b_k(z)\, \lambda^{-k}, \qquad |\lambda| \gg 1. \tag{2.4}$$

The left side is holomorphic at $\lambda = \infty$ (since $h(\lambda) \sim c\lambda$ with $c \neq 0$), so the expansion makes sense. These b_k are the *Faber polynomials* of the exterior coordinate h; they are the natural generalisation of the monomials z^k to the domain Ω. In the simplest case $\Omega = \mathbb{D}$ with $h(\lambda) = \lambda$, one has $b_k(z) = z^k$; for a general Ω, the b_k have corrections coming from the shape of $\partial\Omega$.

We can be more precise: after factoring $c\lambda$ from the denominator, we find $b_0 = 1$ and, for $k \geq 1$, b_k is a polynomial of degree k with leading coefficient c^{-k}. So $(b_k)_{k\geq 0}$ really is a basis of the space of polynomials, and polynomial degrees match indices. This is useful because it lets us rewrite any matrix polynomial $G(z) = \sum_{k=0}^{N} b_k(z)C_k$ uniquely in the Faber basis, with $C_k \in M_m(\mathbb{C})$ in the natural way.

Boundary values and the mixed coefficient

The boundary restrictions $b_k \circ h$ of the Faber polynomials have a remarkably clean Fourier form. There are scalars s_{kj}, indexed by $k, j \geq 1$, such that on the exterior collar one has

$$b_k(h(\mu)) \;=\; \mu^k \;+\; \sum_{j\geq 1} s_{kj}\, \mu^{-j}, \qquad k \geq 1. \tag{2.5}$$

Equivalently, the *mixed correction kernel*

$$s(\lambda, \mu) \;=\; \frac{q(\lambda)}{h(\lambda) - h(\mu)} \;-\; \frac{\lambda}{\lambda - \mu} \;=\; \sum_{k,j\geq 1} s_{kj}\, \lambda^{-k}\mu^{-j} \tag{2.6}$$

collects exactly the difference between the true boundary kernel and the boundary kernel of the unit disk.

The function s has a nice structure. It extends holomorphically across the diagonal $\lambda = \mu$ and across both infinities, so it is regular on a full product collar of the exterior. Its mixed Taylor coefficients decay geometrically:

$$|s_{kj}| \;\leq\; C_\rho\, \rho^{k+j} \qquad (r_0 < \rho < 1,\; k, j \geq 1). \tag{2.7}$$

Consequently the double series converges normally on the exterior product collar, hence absolutely and uniformly on $\mathbb{T}^2$. So every interchange of sum and integral below is legal.

The kernel we will need is

$$K(\lambda, \mu) \;=\; 1 + s(\lambda, \mu) + \overline{s(\lambda, \mu)}. \tag{2.8}$$

Since s has only strictly mixed Fourier modes ($\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$), neither s nor its conjugate contributes to the integral in λ alone or in μ alone. So

$$\int_{\mathbb{T}} K(\lambda, \mu)\, d\tau(\lambda) = 1 \quad \text{(for each fixed } \mu\text{)}, \qquad \int_{\mathbb{T}} K(\lambda, \mu)\, d\tau(\mu) = 1 \quad \text{(for each fixed } \lambda\text{)}.$$

These are the *marginal identities*. They say: K is like the kernel of the identity operator, but with a real, symmetric, nonnegative correction.

The positive kernel theorem

Theorem 2.1 (Exterior Faber coefficients and the positive kernel)**.** *Let Ω be a bounded convex domain with regular real-analytic Jordan boundary. Let h be as in* (**??**)*, and let q, $(b_k)_{k\geq 0}$, and s_{kj} be defined by* (**??**)*,* (**??**)*, and* (**??**)*. Then:*

1. *$b_0 = 1$, and for each $k \geq 1$, b_k is a polynomial of degree k with leading coefficient c^{-k} in* (**??**).

2. *The mixed coefficients s_{kj} satisfy the geometric decay estimate* (**??**).

3. *The kernel K in* (**??**) *is continuous on $\mathbb{T}^2$, nonnegative everywhere, and has integral one in each variable with the other fixed.*

Proof. Part 1 (degree and leading coefficient). Write $h(\lambda) - z = c\lambda \cdot (1 + (c_0 - z)/(c\lambda) + O(\lambda^{-2}))$. For large $|\lambda|$,

$$\frac{q(\lambda)}{h(\lambda) - z} = \frac{c\lambda + O(\lambda^{-1})}{c\lambda} \cdot \left(1 + \frac{c_0 - z}{c\lambda} + \cdots\right)^{-1} = 1 + O(\lambda^{-1}).$$

Expanding the inverse as a geometric series in $(c_0 - z)/(c\lambda)$ produces, for the coefficient of λ^{-k}, a polynomial in z of degree k. Inspecting the largest power of z shows it is $c^{-k}z^k$. So $b_0 = 1$ and $b_k(z) = c^{-k}z^k +$ (lower degree) for $k \geq 1$. In particular the b_k form a basis of the scalar polynomials.

Part 2 (geometric decay and joint extension). Consider

$$H(\lambda, \mu) \;=\; \frac{h(\lambda) - h(\mu)}{\lambda - \mu}. \tag{2.9}$$

Write $u = \lambda^{-1}$, $v = \mu^{-1}$, and $h(\lambda) = c\lambda + g(u)$ with g holomorphic on $|u| < r_0^{-1}$. Then

$$H(\lambda, \mu) = c - uv\,\frac{g(u) - g(v)}{u - v}.$$

The divided difference $(g(u) - g(v))/(u - v)$ is jointly holomorphic in (u, v), with diagonal value $g'(u)$, so this formula covers the diagonal and infinities. Direct differentiation gives

$$s(\lambda, \mu) \;=\; \frac{\lambda\partial_\lambda H}{H} \;=\; -\frac{u\partial_u H}{H}.$$

Since H is nonzero on the reciprocal bidisk $|u|, |v| < r_0^{-1}$ (injectivity of h, and $H \to c$ at infinity), s is holomorphic there and vanishes on both axes $\{u = 0\}$ and $\{v = 0\}$. Its Taylor expansion therefore contains exactly the strictly mixed powers $\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$.

For any $r_0 < \rho < 1$, apply Cauchy estimates on the closed disk $|u|, |v| \leq \rho^{-1}$: there is a constant C_ρ with $|s_{kj}| \leq C_\rho \rho^{k+j}$. Summing the resulting geometric series shows the Laurent expansion (**??**) converges normally (absolutely and uniformly on compacta, including those meeting infinity) throughout the product collar $|\lambda|, |\mu| > r_0$. Restricting to $|\lambda| = |\mu| = 1$ gives absolute uniform convergence on $\mathbb{T}^2$.

Comparing (**??**) and (**??**) coefficient-by-coefficient in λ^{-k}, with μ fixed in the exterior collar, yields (**??**). Indeed, $\lambda/(\lambda - \mu) = \sum_{k \geq 0} \mu^k \lambda^{-k}$, so the difference of the two sides matches $b_k(h(\mu)) - \mu^k$.

Part 3 (positivity of K and marginal identities). For distinct $\lambda, \mu \in \mathbb{T}$, split K as

$$K(\lambda, \mu) \;=\; 2\,\Re\frac{q(\lambda)}{h(\lambda) - h(\mu)} \;+\; \left(1 - \frac{\lambda}{\lambda - \mu} - \frac{\bar\lambda}{\bar\lambda - \bar\mu}\right).$$

On $|\lambda| = |\mu| = 1$, we have $\bar{\lambda} = 1/\lambda$, $\bar{\mu} = 1/\mu$, and one verifies directly that $\lambda/(\lambda-\mu)+\bar{\lambda}/(\bar{\lambda}-\bar{\mu}) = 1$, so the bracketed term vanishes. Hence

$$K(\lambda,\mu) \;=\; 2\,\Re\frac{q(\lambda)}{h(\lambda)-h(\mu)}.$$

Now $\Re(q/(h(\lambda)-h(\mu))) = |q|^2/\Re(\overline{q}(h(\lambda)-h(\mu)))$ where $|q|^2 > 0$, and the denominator is nonnegative by (**??**) with $z = h(\mu)$. So $K(\lambda,\mu) \geq 0$ for distinct λ, μ, and continuity gives the same on the diagonal.

For the marginal identities: every term in s has Fourier mode $\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$, so integrating against $d\tau(\lambda)$ picks out $k = 0$, which never occurs; and similarly for the conjugate. Hence $\int_{\mathbb{T}} s(\lambda,\mu)d\tau(\lambda) = \int_{\mathbb{T}} \overline{s(\lambda,\mu)}d\tau(\lambda) = 0$, so $\int_{\mathbb{T}} K(\lambda,\mu)d\tau(\lambda) = 1$. Symmetric argument in μ. □

Identification of b_k as the Faber polynomial

Let $\Phi = h^{-1}$ denote the inverse of h on the exterior collar $\mathbb{C}\setminus\overline{\Omega} \to \{\lambda : |\lambda| > 1\}$. Then (**??**), which continues to hold throughout the exterior collar, gives

$$b_k(z) - \Phi(z)^k = O(z^{-1}), \qquad z \to \infty.$$

So b_k is the polynomial part of Φ^k – the classical *Faber polynomial* associated with the exterior coordinate h (see [**?**, **?**] for background). This is why the notation b_k is standard.

The mixed coefficients s_{kj} are closely related to the differentiated *Grunsky coefficients* of the exterior map Φ; the classical normalization gives a weighted coefficient inequality [**?**, eq. (1.1)–(1.3)]. The positivity estimate we need below is a consequence of convexity and the marginal identities already proved in Theorem **??**; no extra Grunsky estimate is required.

3 The coefficient norm estimate

The positive kernel K from Theorem **??** is the entire source of the constant 2 in the main theorem. This section extracts one clean inequality from it. Fix any dimension m and any finite-family of coefficient matrices $C_0, C_1, \ldots, C_N \in M_m(\mathbb{C})$, and put

$$G(z) \;=\; \sum_{k=0}^{N} b_k(z)\, C_k.$$

That is, G is an $M_m(\mathbb{C})$-valued polynomial written in the Faber basis of Ω. We want to bound the Hilbert–Schmidt L^2-norm of $G \circ h$ on the unit circle from above and below by the coefficient Hilbert–Schmidt norms $\|C_k\|_2$.

The operator T defined by K

Consider the bounded operator

$$(Tu)(\mu) \;=\; \int_{\mathbb{T}} K(\lambda,\mu)\, u(\lambda)\, d\tau(\lambda), \qquad u \in L^2(\mathbb{T};\mathcal{H}), \tag{3.1}$$

where $\mathcal{H}$ is any separable Hilbert space (in practice $\mathcal{H} = M_m(\mathbb{C})$ with the Hilbert–Schmidt inner product).

Lemma 3.1. *The operator T extends to a contraction of $L^2(\mathbb{T};\mathcal{H})$.*

Proof. Nonnegativity $K \geq 0$ and the marginal identity $\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\lambda) = 1$ together give, for each fixed μ, by Cauchy–Schwarz,

$$\|(Tu)(\mu)\|^2 \;\leq\; \left(\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\lambda)\right)\left(\int_{\mathbb{T}} K(\lambda,\mu)\,\|u(\lambda)\|^2\,d\tau(\lambda)\right) \;=\; \int_{\mathbb{T}} K(\lambda,\mu)\,\|u(\lambda)\|^2\,d\tau(\lambda).$$

Integrate over μ and use the other marginal identity:

$$\|Tu\|_{L^2}^2 \;\leq\; \int_{\mathbb{T}}\left(\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\mu)\right)\|u(\lambda)\|^2\,d\tau(\lambda) \;=\; \|u\|_{L^2}^2.$$

□

What T does on monomials

For a vector-valued function of the form $u(\lambda) = \sum_{k=1}^{N} C_k\lambda^k$, where each $C_k \in M_m(\mathbb{C})$, direct substitution in (**??**) shows that only the s summand of K contributes: the term 1 in K gives a zero integral (nonzero frequencies $\lambda^1,\dots,\lambda^N$), and the term $\overline{s}$ gives a zero integral because $\overline{s(\lambda,\mu)} = \sum_{k,j\geq 1}\overline{s_{kj}}\lambda^k\mu^j$, whose product with $C_k\lambda^k$ has only nonnegative powers of λ. So

$$(Tu)(\mu) \;=\; \sum_{j\geq 1}\Big(\sum_{k=1}^{N} s_{kj}\,C_k\Big)\,\mu^{-j}. \tag{3.2}$$

The normal convergence in Theorem **??** makes all interchanges legal. Because the μ^{-j} are orthonormal in $L^2(\mathbb{T})$,

$$\|Tu\|_{L^2}^2 \;=\; \sum_{j\geq 1}\Big\|\sum_{k=1}^{N} s_{kj}\,C_k\Big\|_2^2,$$

where $\|\cdot\|_2$ denotes the Hilbert–Schmidt norm on $M_m(\mathbb{C})$. Combined with Lemma **??**,

$$\sum_{j\geq 1}\Big\|\sum_{k=1}^{N} s_{kj}\,C_k\Big\|_2^2 \;\leq\; \sum_{k=1}^{N}\|C_k\|_2^2. \tag{3.3}$$

The two-sided coefficient inequality

Now observe that on the boundary circle, by (**??**),

$$G(h(\mu)) \;=\; \sum_{k=0}^{N} b_k(h(\mu))C_k \;=\; C_0 + \sum_{k=1}^{N} C_k\mu^k + \sum_{k=1}^{N}\left(\sum_{j\geq 1} s_{kj}\,\mu^{-j}\right)C_k \;=\; C_0 + u(\mu) + (Tu)(\mu).$$

The three summands C_0, u, and Tu have disjoint Fourier supports (constant, positive, negative). Parseval's identity applies:

$$\int_{\mathbb{T}}\|G(h(\mu))\|_2^2\,d\tau(\mu) \;=\; \|C_0\|_2^2 + \sum_{k=1}^{N}\|C_k\|_2^2 + \|Tu\|_{L^2}^2.$$

Combine this identity with (**??**). The case gives

$$\|C_0\|_2^2 + \sum_{k=1}^{N}\|C_k\|_2^2 \;\leq\; \int_{\mathbb{T}}\|G(h(\mu))\|_2^2\,d\tau(\mu) \;\leq\; \|C_0\|_2^2 + 2\sum_{k=1}^{N}\|C_k\|_2^2.$$

This is the key estimate. In words: passing from a coefficient array $(C_0, C_1, \dots)$ to the boundary values of G can enlarge the coefficient Hilbert–Schmidt norm by at most a universal factor of 2, and never shrinks it.

Proposition 3.2 (The coefficient norm estimate)**.** *For every positive integer m, every integer $N \geq 0$, and every $M_m(\mathbb{C})$-valued polynomial $G(z) = \sum_{k=0}^{N} b_k(z)\, C_k$ in the Faber basis of Ω, one has*

$$\sum_{k=0}^{N} \|C_k\|_2^2 \;\leq\; \int_{\mathbb{T}} \|G(h(\mu))\|_2^2 \, d\tau(\mu) \;\leq\; \|C_0\|_2^2 + 2 \sum_{k=1}^{N} \|C_k\|_2^2. \tag{3.4}$$

Proof. This is exactly the combined identity above. The lower bound uses only $\|Tu\|^2 \geq 0$. The upper bound uses Lemma **??** to shoehorn $\|Tu\|_{L^2} \leq \sum_k \|C_k\|_2$ into the Parseval identity. □

A note on the literature

For comparison with [**?**, Proposition 3.1], the companion paper's exterior map G and variables t, w correspond to h, λ, μ, respectively. The mixed correction satisfies $s(\lambda, \mu) = B(\mu, \lambda)$, so $s_{kj} = \beta_{jk}$ in that paper's notation. Our formula (**??**) uses this orientation, and the proof above does not require symmetry of the kernel.

4 The resolvent and the double-layer positivity

From now on the matrix $A \in M_n(\mathbb{C})$ is fixed, with $W(A) \subset \Omega$. We will also fix the exterior coordinate h of Ω from section **??**. The purpose of this section is to define the *exterior resolvent* and show that it has a positive Hermitian part.

The resolvent function

Define, on the exterior circle $\mathbb{T}$,

$$R(\lambda) \;=\; q(\lambda)\, (h(\lambda) I - A)^{-1} \;\in\; M_n(\mathbb{C}). \tag{4.1}$$

Here $h(\lambda)I - A$ is invertible whenever $h(\lambda) \notin \sigma(A)$. Since $\sigma(A) \subset W(A) \subset \Omega$ (compact) and the circle $\mathbb{T}$ parametrises $\partial\Omega$ (compact, disjoint from $\sigma(A)$), the resolvent exists and is continuous on $\mathbb{T}$, and extends holomorphically to a collar $\{|\lambda| > r_A\}$ for some $r_0 < r_A < 1$ with r_A closer to one than r_0.

Theorem 4.1 (Resolvent expansion and double-layer positivity)**.** *Let $\Omega, h, q, (b_k)$ be as above. For every $n \geq 1$ and every $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$, the following hold.*

1. Expansion. *R has the Laurent series*

$$R(\lambda) \;=\; \sum_{k \geq 0} b_k(A)\, \lambda^{-k}, \qquad |\lambda| \geq 1, \tag{4.2}$$

   converging absolutely and uniformly on $\mathbb{T}$. The coefficient matrices $b_k(A)$ decay geometrically: there are constants $C < \infty$ and $0 < \theta < 1$ (depending on A and h) with $\|b_k(A)\| \leq C\theta^k$ for all $k \geq 0$.

2. Means. *$\int_{\mathbb{T}} R(\lambda)\, d\tau(\lambda) = I$ and $\int_{\mathbb{T}} (R(\lambda) + R(\lambda)^*)\, d\tau(\lambda) = 2I$.*

3. Positivity. *On the exterior circle,* $R(\lambda) + R(\lambda)^* \succeq 0$.

Proof. Part 1 (expansion and decay). Every eigenvalue of A belongs to $W(A) \subset \Omega$, by testing the corresponding unit eigenvector. Hence the resolvent $(h(\lambda)I - A)^{-1}$ is holomorphic on and outside $\mathbb{T}$, including a small exterior collar. Expanding at ∞ in λ gives a Laurent series in λ^{-1}, and comparing with (**??**) with z replaced by A shows that the coefficient of λ^{-k} is exactly $b_k(A)$. The series converges absolutely and uniformly on $\mathbb{T}$ by compactness of the exterior collar. To get the geometric bound, pick $r_A < \theta < 1$ with R holomorphic on $|\lambda| > r_A$, apply Cauchy estimates on the circle $|\lambda| = \theta^{-1}$.

Part 2 (means). The constant coefficient $b_0(A) = b_0(z)\big|_{z=A} = I$ (Theorem **??**, part 1). So by (**??**) and orthogonality of the modes λ^{-k}, $\int_{\mathbb{T}} R\, d\tau = b_0(A) = I$. Taking adjoints and adding gives $\int_{\mathbb{T}}(R + R^*)\, d\tau = 2I$.

Part 3 (positivity). Let $B(\lambda) = h(\lambda)I - A$ so that $R = qB^{-1}$. Then $R^* = \overline{q}\,(B^*)^{-1}$ and

$$R + R^* \;=\; q\,B^{-1} + \overline{q}\,(B^*)^{-1} \;=\; (B^*)^{-1}\,(\overline{q}\,B + q\,B^*)\,B^{-1}.$$

Since $(B^*)^{-1}$ and B^{-1} are adjoints of each other, they give a congruence that preserves positivity. So it suffices to show that $\overline{q(\lambda)}\,B(\lambda) + q(\lambda)\,B(\lambda)^* \succeq 0$.

For any unit vector $v \in \mathbb{C}^n$, the quadratic form on the left at v is

$$v^*\,(\overline{q}\,B + q\,B^*)\,v \;=\; \overline{q}\,v^*Bv + q\,v^*B^*v \;=\; 2\Re\Big(\overline{q(\lambda)}\,v^*B(\lambda)v\Big)\,.$$

Substituting $B = h(\lambda)I - A$:

$$v^*Bv \;=\; h(\lambda) - v^*Av.$$

So the quadratic form is $\Re\Big(\overline{q(\lambda)}(h(\lambda) - v^*Av)\Big)$. By the support inequality (**??**) with $z = v^*Av \in W(A) \subset \overline{\Omega}$, this is ≥ 0. The congruence argument finishes the proof. □

5 The main estimate on an analytic convex domain

Now the payoff. In this section $\Omega, h, q, (b_k)$ are as in Section **??**, $A \in M_n(\mathbb{C})$ satisfies $W(A) \subset \Omega$, and $R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}$ is the exterior resolvent from Theorem **??**.

Statement and strategy

Theorem 5.1 (Bound on an analytic convex domain)**.** *Let $n, m \geq 1$. Let $\Omega \subset \mathbb{C}$ be bounded, open, convex, with regular real-analytic Jordan boundary, and let $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$. Let F be an $M_m(\mathbb{C})$-valued polynomial with $\|F(z)\| \leq 1$ on $\partial\Omega$ (equivalently, on $\overline{\Omega}$, since F is a polynomial and the boundary controls the closure). Then*

$$\|F[A]\| \;\leq\; 2.$$

The proof is a bookkeeping exercise with three moving pieces:

- Top singular vectors x, y of $F[A]$ that turn the norm into a *scalar* $y^*F[A]x = \|F[A]\|$.
- The two products FG and GF that turn $x^*G[A]x$ and $y^*G[A]y$ into the same scalar $y^*(\cdot)[A]x$ for two different choices of G.
- The two-by-two block $R + R^* \succeq 0$ that lets positivity compare the three fields $P + P^*$, $Q + Q^*$, and $M + M^*$ to each other.

Each piece contributes one inequality, and together they force $\|F[A]\| \leq 2$.

Setup: singular vectors and block notation

Put $\gamma = \|F[A]\|$. If $\gamma = 0$, there is nothing to prove, so assume $\gamma > 0$. Choose unit vectors $x, y \in \mathbb{C}^n \otimes \mathbb{C}^m$ such that

$$F[A]x \;=\; \gamma y, \qquad F[A]^* y \;=\; \gamma x. \tag{5.1}$$

Such a pair exists by the finite-dimensional singular value decomposition.

Identify $\mathbb{C}^n \otimes \mathbb{C}^m \cong (\mathbb{C}^n)^m = \mathbb{C}^{nm}$ via the standard basis $e_1, \dots, e_m$ of $\mathbb{C}^m$. Any vector $x \in \mathbb{C}^n \otimes \mathbb{C}^m$ can be written uniquely as $x = \sum_{i=1}^m Xe_i \otimes e_i$ for an $n \times m$ matrix X. Similarly, write $y = \sum_{i=1}^m Ye_i \otimes e_i$ with Y an $n \times m$ matrix. The unit-length conditions become

$$\|X\|_2^2 = \|x\|^2 = 1, \qquad \|Y\|_2^2 = \|y\|^2 = 1$$

(here and below $\|X\|_2^2 = \operatorname{tr}(X^*X)$).

Define the $m \times m$ matrix functions

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY. \tag{5.2}$$

Their Laurent coefficients in the variable λ^{-1} will be denoted with subscripts: $M_k = Y^*b_k(A)X$, $P_k = X^*b_k(A)X$, $Q_k = Y^*b_k(A)Y$, $L_k = X^*b_k(A)Y$. Since $\|b_k(A)\| \le C\theta^k$ (Theorem **??**), all four series converge absolutely and uniformly on $\mathbb{T}$, and all four are square-summable in the Hilbert-Schmidt norm.

Important bookkeeping at $k = 0$. Since $b_0(A) = I$,

$$P_0 = X^*X, \quad Q_0 = Y^*Y, \quad L_0^* = M_0. \tag{5.3}$$

In particular P_0 and Q_0 are Hermitian and positive semidefinite.

Finally, set

$$a^2 \;=\; \sum_{k\ge 0} \|M_k\|_2^2, \qquad \text{finite and } a > 0. \tag{5.4}$$

The finiteness is from the geometric decay of $b_k(A)$. The positivity $a > 0$ follows from applying the cross-functional bound below to $G = F$: the value $y^*F[A]x = \gamma > 0$ cannot be achieved by a Cauchy-Schwarz bound with a zero right-hand side.

Step A. The off-diagonal functional

For any $M_m(\mathbb{C})$-valued polynomial $G(z) = \sum_{k=0}^N b_k(z)C_k$ written in the Faber basis, direct expansion in the tensor product gives

$$y^*G[A]x \;=\; \sum_{k=0}^{N} \sum_{i,j=1}^{m} (M_k)_{ij}\,(C_k)_{ij}. \tag{5.5}$$

Let us verify (**??**) once (it is used three more times). Since $x = \sum_j Xe_j \otimes e_j$ and $G[A] = \sum_k b_k(A) \otimes C_k$,

$$G[A]x \;=\; \sum_{k,j} \big(b_k(A)Xe_j\big) \otimes \big(C_k e_j\big).$$

Pairing with $y = \sum_i Ye_i \otimes e_i$:

$$y^*G[A]x \;=\; \sum_{k,i,j} (Ye_i)^* b_k(A)(Xe_j) \cdot e_i^* C_k e_j \;=\; \sum_{k,i,j} (M_k)_{ij}\,(C_k)_{ij},$$

as claimed. Equation (**??**) is a scalar bilinear pairing of the coefficient arrays $(M_k)_{ij}$ and $(C_k)_{ij}$.

Cauchy-Schwarz on (**??**) gives

$$|y^*G[A]x| \;\le\; \left(\sum_{k\ge 0}\|M_k\|_2^2\right)^{1/2}\left(\sum_{k=0}^{N}\|C_k\|_2^2\right)^{1/2} \;=\; a\left(\sum_{k=0}^{N}\|C_k\|_2^2\right)^{1/2}. \tag{5.6}$$

Plugging in $G = F$ and using $y^*F[A]x = \gamma$ from (**??**) shows that $a > 0$.

Step B. Two ordered tests

Polynomial evaluation with matrix coefficients respects multiplication in the given order. If $F(z) = \sum_i D_i z^i$ and $G(z) = \sum_j E_j z^j$, then

$$(FG)[A] \;=\; \sum_{i,j} A^{i+j}\otimes D_iE_j \;=\; F[A]G[A].$$

Similarly $(GF)[A] = G[A]F[A]$ when factors are in the reversed order. Multiplying (**??**) on the left by suitable pieces,

$$y^*(FG)[A]x \;=\; y^*F[A]G[A]x \;=\; \gamma\, x^*G[A]x, \qquad y^*(GF)[A]x \;=\; \gamma\, y^*G[A]y. \tag{5.7}$$

So the diagonal functionals $x^*G[A]x$ and $y^*G[A]y$ can be turned into the off-diagonal functional by ordering the two factors differently.

Write the two products in the Faber basis:

$$FG \;=\; \sum_{k\ge 0} b_kU_k, \qquad GF \;=\; \sum_{k\ge 0} b_kV_k,$$

with finitely many nonzero coefficient matrices $U_k, V_k \in M_m(\mathbb{C})$. Define the boundary norm

$$\|H\|_\partial^2 \;=\; \int_{\mathbb{T}} \|H(h(\mu))\|_2^2\, d\tau(\mu), \qquad H \in M_m(\mathbb{C})[z].$$

Because $\|F(z)\| \le 1$ on the boundary, submultiplicativity of the pointwise operator norm gives

$$\|FG\|_\partial \;\le\; \|F\|_\partial\,\|G\|_\partial \;\le\; \|G\|_\partial, \qquad \|GF\|_\partial \;\le\; \|G\|_\partial.$$

(The two inequalities $\|F(z)G(z)\|_2 \le \|F(z)\|\|G(z)\|_2$ and $\|G(z)F(z)\|_2 \le \|G(z)\|_2\|F(z)\|$ are used in their respective orders.)

Apply (**??**) to FG and GF, and use (**??**), the lower coefficient estimate from Proposition **??**, and the boundary multiplier bounds:

$$\gamma\,|x^*G[A]x| = |y^*(FG)[A]x| \le a\left(\sum_k \|U_k\|_2^2\right)^{1/2} \le a\|FG\|_\partial \le a\|G\|_\partial,$$

$$\gamma\,|y^*G[A]y| = |y^*(GF)[A]x| \le a\left(\sum_k \|V_k\|_2^2\right)^{1/2} \le a\|GF\|_\partial \le a\|G\|_\partial.$$

The upper coefficient estimate for G in Proposition **??** turns $\|G\|_\partial$ into a norm of the coefficient array: $\|G\|_\partial \le (\|C_0\|_2^2 + 2\sum_{k\ge 1}\|C_k\|_2^2)^{1/2}$. Therefore

$$\max\{|x^*G[A]x|, |y^*G[A]y|\} \;\le\; \frac{a}{\gamma}\left(\|C_0\|_2^2 \;+\; 2\sum_{k\ge 1}\|C_k\|_2^2\right)^{1/2}. \tag{5.8}$$

This is the crucial estimate that produces the weight of two.

Step C. Weighted dual norms

The functional $x^*G[A]x$ pairs C_k against P_k (repeat the calculation of (**??**), but with y replaced by x; the coefficient array is $P_k = X^*b_k(A)X$). Similarly $y^*G[A]y$ pairs C_k against Q_k. So (**??**) constrains the size of the sequences (P_k) and (Q_k), with weights that reverse the ones in (**??**).

Concretely, for each $N \geq 0$ set

$$d_N^{(P)} \;=\; \|P_0\|_2^2 \;+\; \tfrac{1}{2}\sum_{k=1}^{N} \|P_k\|_2^2.$$

Test (**??**) with the coefficient sequence

$$C_0 = \overline{P_0}, \qquad C_k = \overline{P_k}/2 \;(1 \leq k \leq N), \qquad C_k = 0 \;(k > N),$$

where the bar denotes entrywise complex conjugation. The scalar paired value is exactly $d_N^{(P)}$. The weighted norm on the right is $\|C_0\|_2^2 + 2\sum_{k=1}^{N} \|C_k\|_2^2 = \|P_0\|_2^2 + \tfrac{1}{2}\sum_{k=1}^{N} \|P_k\|_2^2 = d_N^{(P)}$. So

$$d_N^{(P)} \;\leq\; \frac{a}{\gamma}\,(d_N^{(P)})^{1/2}.$$

If $d_N^{(P)} = 0$ this is trivial. Otherwise divide and square: $d_N^{(P)} \leq a^2/\gamma^2$. Letting $N \to \infty$:

$$\|P_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|P_k\|_2^2 \;\leq\; \frac{a^2}{\gamma^2}, \qquad \|Q_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|Q_k\|_2^2 \;\leq\; \frac{a^2}{\gamma^2}. \tag{5.9}$$

The same finite test with Q_k in place of P_k gives the second bound.

Step D. Boundary L^2 norms of the blocks

Now compute the square-integrated Hilbert-Schmidt norms of the three extreme matrix fields $P + P^*$, $Q + Q^*$, and $M + L^*$ on the circle.

Diagonal blocks. The function $P + P^*$ has Laurent expansion

$$(P + P^*)(\lambda) \;=\; 2P_0 \;+\; \sum_{k\geq 1} P_k\,\lambda^{-k} \;+\; \sum_{k\geq 1} P_k^*\,\lambda^{k}$$

where we used $P_0^* = P_0$ and $\overline{\lambda}^k = \lambda^{-k}$ on $|\lambda| = 1$. The three groups (constant, negative powers, positive powers) have disjoint Fourier support, so Parseval gives

$$\int_{\mathbb{T}} \|P + P^*\|_2^2\, d\tau(\lambda) \;=\; \|2P_0\|_2^2 \;+\; \sum_{k\geq 1} \|P_k\|_2^2 \;+\; \sum_{k\geq 1} \|P_k^*\|_2^2 \;=\; 4\|P_0\|_2^2 \;+\; 2\sum_{k\geq 1} \|P_k\|_2^2 \;=\; 4d_\infty^{(P)},$$

where $d_\infty^{(P)} = \|P_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|P_k\|_2^2$ is the limit of the sequence $d_N^{(P)}$ from Step C. Combined with (**??**),

$$\int_{\mathbb{T}} \|P + P^*\|_2^2\, d\tau \;\leq\; 4a^2/\gamma^2, \qquad \int_{\mathbb{T}} \|Q + Q^*\|_2^2\, d\tau \;\leq\; 4a^2/\gamma^2. \tag{5.10}$$

The second inequality follows from the second half of (**??**) by the same calculation with Q in place of P.

Cross block. The function $M + L^*$ has Laurent expansion

$$(M + L^*)(\lambda) \;=\; \sum_{k\geq 0} M_k\, \lambda^{-k} \;+\; \sum_{k\geq 0} L_k^*\, \lambda^k.$$

The constant coefficient is $M_0 + L_0^* = 2M_0$, using $L_0^* = M_0$ from (**??**). Grouping by Fourier mode,

$$\int_{\mathbb{T}} \|M + L^*\|_2^2\, d\tau \;=\; \|2M_0\|_2^2 \;+\; \sum_{k\geq 1} \|M_k\|_2^2 \;+\; \sum_{k\geq 1} \|L_k^*\|_2^2.$$

Since $\|L_k^*\|_2 = \|L_k\|_2$ (the Hilbert-Schmidt norm is invariant under adjoint) and every term is nonnegative,

$$\int_{\mathbb{T}} \|M + L^*\|_2^2\, d\tau \;\geq\; 4\|M_0\|_2^2 \;+\; \sum_{k\geq 1} \|M_k\|_2^2 \;\geq\; \sum_{k\geq 0} \|M_k\|_2^2 \;=\; a^2. \tag{5.11}$$

The final inequality holds because the coefficient of $4\|M_0\|_2^2$ exceeds the coefficient of $\|M_0\|_2^2$; dropping the $\|L_k\|_2^2$ terms only makes the left smaller.

Step E. Positive block comparison

The proof now closes with linear algebra.

From Theorem **??**, $R(\lambda) + R(\lambda)^* \succeq 0$ for every $\lambda \in \mathbb{T}$. Compress by the $n \times 2m$ matrix $[X\ Y]$. The $2m \times 2m$ matrix of blocks

$$\Delta(\lambda) \;=\; \begin{pmatrix} P + P^* & L + M^* \\ M + L^* & Q + Q^* \end{pmatrix}(\lambda) \;=\; \begin{pmatrix} X \\ Y \end{pmatrix}^* (R + R^*)(\lambda) \begin{pmatrix} X & Y \end{pmatrix} \;\succeq\; 0$$

is positive semidefinite for every λ. A compression of a positive semidefinite matrix is positive semidefinite, so this is immediate.

Let $J = \operatorname{diag}(I_m, -I_m)$. If $\Delta \succeq 0$, then $J\Delta J \succeq 0$ too (the conjugation by J keeps positivity). If two matrices S and T are positive semidefinite, then $\operatorname{tr}(ST) \geq 0$: writing $S = AA^*$ and $T = BB^*$, $\operatorname{tr}(ST) = \operatorname{tr}(AA^*BB^*) = \operatorname{tr}((A^*B)^*(A^*B)) = \|A^*B\|_2^2 \geq 0$. So

$$0 \;\leq\; \operatorname{tr}(\Delta(J\Delta J)) \;=\; \operatorname{tr}((\Delta J)^2).$$

Compute $(\Delta J)^2$ directly. $\Delta J = \begin{pmatrix} P + P^* & -(L + M^*) \\ M + L^* & -(Q + Q^*) \end{pmatrix}$. Then

$$\operatorname{tr}\big((\Delta J)^2\big) \;=\; \operatorname{tr}\big((P+P^*)^2\big) \;-\; \operatorname{tr}\big((L+M^*)(M+L^*)\big) \;-\; \operatorname{tr}\big((M+L^*)(L+M^*)\big) \;+\; \operatorname{tr}\big((Q+Q^*)^2\big).$$

Since $\operatorname{tr}((AB)^*) = \overline{\operatorname{tr}(AB)}$ but also $(L + M^*)(M + L^*)$ and its adjoint $(M + L^*)(L + M^*)$ have the same trace (they are adjoints, and their traces are complex conjugates; both equal $\operatorname{tr}(AA^*)$ with $A = L + M^*$, which is real nonnegative), the two middle terms sum to

$$-\operatorname{tr}\big((L + M^*)(M + L^*) + (M + L^*)(L + M^*)\big) \;=\; -2\operatorname{tr}(L + M^*)^*(L + M^*) \;=\; -2\|M + L^*\|_2^2.$$

Putting it all together:

$$0 \;\leq\; \|P + P^*\|_2^2 \;+\; \|Q + Q^*\|_2^2 \;-\; 2\|M + L^*\|_2^2. \tag{5.12}$$

This holds for every $\lambda \in \mathbb{T}$.

Integrate (**??**) over $\mu \in \mathbb{T}$, use (**??**) for the two diagonal terms and (**??**) for the cross term:

$$0 \;\le\; 4a^2/\gamma^2 + 4a^2/\gamma^2 \;-\; 2a^2 \;=\; 8a^2/\gamma^2 \;-\; 2a^2.$$

So

$$2a^2 \;\le\; 8a^2/\gamma^2.$$

Since $a > 0$, divide by a^2: $\gamma^2 \le 4$, hence $\gamma \le 2$.

This finishes the proof that $\|F[A]\| \le 2$. □

Remarks on the argument

1. *Where the* 2 *comes from.* The factor of two in the upper coefficient estimate (**??**) is multiplied through Step E until it becomes $\gamma^2 = 4$. Removing the factor of two would give $\gamma^2 \le 2$, and that is false in general (the sharpness example in Section **??** has $\gamma = 2$).

2. *Where convexity enters.* The only place convexity is used is in the support inequality (**??**), which in turn gives the positivity of K (Theorem **??**) and the positivity of $R + R^*$ (Theorem **??**). If Ω were not convex, both positivity statements could fail and the argument would not go through.

3. *Why scalar proofs do not suffice.* In a scalar argument ($m = 1$), one would use $FG = GF$ freely. Here, the two products FG and GF are genuinely different $M_m(\mathbb{C})$-valued polynomials. The whole point of Step B is to use these two different products separately, obtaining bounds on $x^*G[A]x$ and $y^*G[A]y$ that share the same cross-functional a.

6 Enclosing domains and the general case

We have proved $\|F[A]\| \le 2$ under the hypothesis $W(A) \subset \Omega$ with Ω an analytic convex domain. We now remove that hypothesis: for an arbitrary compact convex $W(A)$ we construct analytic convex neighborhoods Ω_j shrinking to $W(A)$, apply the previous section, and pass to the limit.

Analytic convex outer approximation

Lemma 6.1 (Analytic convex outer approximation)**.** *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. There exist bounded open convex domains $\Omega_j \subset \mathbb{C}$ with regular real-analytic Jordan boundaries such that $K \subset \Omega_j$ for every j and*

$$\sup_{z \in \overline{\Omega}_j} \operatorname{dist}(z, K) \;\xrightarrow[j\to\infty]{}\; 0.$$

Proof. The construction is a smooth convex interpolation between K and a small thickening. Fix $0 < \delta < 1$. Choose a finite δ-net D_δ of unit directions that includes $1, -1, i, -i$. Let $s_K(\nu) = \max_{z \in K} \Re(\overline{\nu} z)$ be the support function of K. Choose $t_\delta > 0$ small enough that $|D_\delta| e^{-t_\delta \delta} < 1$.

Define the smooth, strictly convex function

$$\Phi_\delta(z) \;=\; \sum_{\nu \in D_\delta} \exp\bigl(t_\delta \,(\Re(\overline{\nu} z) - s_K(\nu) - \delta)\bigr)$$

and set $\Omega_\delta = \{z \in \mathbb{C} : \Phi_\delta(z) < 1\}$.

(i) $K \subset \Omega_\delta$. For $z \in K$ and any ν, $\Re(\overline{\nu} z) \le s_K(\nu)$, so each exponential is at most $e^{-t_\delta \delta}$, and the sum is at most $|D_\delta| e^{-t_\delta \delta} < 1$.

(ii) $\overline{\Omega}_\delta$ lies in a fixed rectangle. If $\Phi_\delta(z) \le 1$, then every summand is ≤ 1 (since each is nonnegative), hence $\Re(\overline{\nu} z) \le s_K(\nu) + \delta$ for every $\nu \in D_\delta$. In particular, for $\nu = 1$ and $\nu = -1$, we get $\Re z$ in $[\min_{w \in K} \Re w - \delta, \max_{w \in K} \Re w + \delta]$; and similarly the imaginary part. So all $\overline{\Omega}_\delta$ for $0 < \delta < 1$ sit inside one fixed bounded rectangle $R \subset \mathbb{C}$.

(iii) Φ_δ is strictly convex with positive definite Hessian. The Hessian is a sum of positive multiples of $\nu\nu^{\mathsf{T}}$ (outer products), where ν ranges over D_δ. Each $\nu\nu^{\mathsf{T}}$ is positive semidefinite; the four directions $1, -1, i, -i$ together span $\mathbb{R}^2$, so the sum is strictly positive definite. Hence Φ_δ is strictly convex and real analytic.

(iv) Level set is regular and a Jordan curve. If z_0 were a critical point of Φ_δ with $\Phi_\delta(z_0) = 1$, then by convexity Φ_δ would attain a global minimum at z_0, contradicting $\Phi_\delta(\text{any } z \in K) < 1$. So $\nabla \Phi_\delta \neq 0$ on $\Phi_\delta^{-1}(1)$. The implicit function theorem makes that level a regular real-analytic curve. Pick an interior point $w_0 \in \Omega_\delta$ (any point of K will do). Each ray from w_0 crosses $\partial\Omega_\delta$ exactly once, because Φ_δ is strictly convex and bounded. Radial projection gives a continuous bijection $\partial\Omega_\delta \to \mathbf{S}^1$, so $\partial\Omega_\delta$ is a Jordan curve and $\overline{\Omega}_\delta = \{\Phi_\delta \le 1\}$.

(v) Distance to K. For $z \in \overline{\Omega}_\delta$, let $w \in K$ be the unique nearest point of K and put $d = |z - w|$. If $z \in K$, then $d = 0$. Otherwise $d > 0$; set $\nu = (z - w)/d$, a unit vector. Choose $\mu \in D_\delta$ with $|\mu - \nu| \le \delta$, and choose $v \in K$ attaining $s_K(\mu) = \Re(\overline{\mu} v)$.

Using minimality of w, for every $w' \in K$ we have $\Re(\overline{\nu}(w' - w)) \le 0$ (the real part of the derivative of $|z - w|^2$ along $w + t(w' - w)$ at $t = 0$). In particular $\Re(\overline{\nu}(v - w)) \le 0$ and $\Re(\overline{\mu}(w - v)) \le 0$.

Now compute:

- Since $|\mu - \nu| \le \delta$, $\Re(\overline{\mu}\nu) \ge 1 - \delta$ and hence $\Re(\overline{\mu}(z - w)) = d\Re(\overline{\mu}\nu) \ge (1 - \delta)\, d$.
- Since $\Re(\overline{\mu}(v - w)) \ge 0$ (support inequality, since v maximizes $\Re(\overline{\mu} \cdot)$ on K) and since $\Phi_\delta(z) \le 1$ forces $\Re(\overline{\mu} z) \le \Re(\overline{\mu} v) + \delta$, we get $\Re(\overline{\mu}(z - v)) \le \delta$.
- From $\Re(\overline{\nu}(v - w)) \le 0$, together with $|\mu - \nu| \le \delta$, we get $\Re(\overline{\mu}(v - w)) \le \delta |v - w|$.

Hence

$$(1 - \delta)\, d \;\le\; \Re(\overline{\mu}(z - w)) \;=\; \Re(\overline{\mu}(z - v)) + \Re(\overline{\mu}(v - w)) \;\le\; \delta + \delta\, |v - w| \;\le\; \delta(1 + \operatorname{diam} K).$$

Points of K have distance zero, so

$$\sup_{z \in \overline{\Omega}_\delta} \operatorname{dist}(z, K) \;\le\; \frac{\delta(1 + \operatorname{diam} K)}{1 - \delta}.$$

Taking $\delta = \delta_j = 1/(j + 1)$ and $\Omega_j = \Omega_{\delta_j}$ completes the construction. □

Proof of the main theorem

Proof of Theorem **??**. First recall the Toeplitz–Hausdorff convexity theorem [**?**, **?**]: $W(A)$ is convex.

Compactly. $W(A) = \{u^* A u : u^* u = 1\}$ is the image of the unit sphere (compact) under a continuous map, hence is compact.

Convexly. Take any $u, v \in \mathbb{C}^n$ with $|u| = |v| = 1$. We want to show that the segment $[u^* A u, v^* A v]$ lies in $W(A)$. Let $V = \operatorname{span}\{u, v\}$, of complex dimension 1 or 2.

If $\dim V = 1$, then $v = e^{i\theta} u$ and $v^* A v = u^* A u$, so the segment is trivial.

If $\dim V = 2$, work in an orthonormal basis $\{e_1, e_2\}$ of V, and let B be the 2×2 matrix of $P_V A|_V$ in this basis. For a unit vector $x = x_1 e_1 + x_2 e_2$, writing $X = 2\Re(\overline{x_1} x_2)$, $Y = 2\Im(\overline{x_1} x_2)$, $Z = |x_1|^2 - |x_2|^2$, one has $X^2 + Y^2 + Z^2 = 1$ (because $4|x_1 x_2|^2 + (|x_1|^2 - |x_2|^2)^2 = (|x_1|^2 + |x_2|^2)^2 = 1$), so (X, Y, Z) ranges over the unit sphere $\mathbf{S}^2 \subset \mathbb{R}^3$, and $x^* B x$ is a real-linear function $f(X, Y, Z)$ into $\mathbf{C} \cong \mathbb{R}^2$. The linear part of f has a one-dimensional kernel. Every point of the unit ball can be pushed along the kernel direction onto the sphere without changing f, so $f(\mathbf{S}^2) = f(\overline{\mathbb{B}^3})$. The image of a ball under a linear map is convex, so $f(\mathbf{S}^2)$ is convex. In particular, the segment $[u^* A u, v^* A v] \subset f(\mathbf{S}^2) \subset W(A)$.

So $W(A)$ is compact and convex.

Enclosing. Apply Lemma **??** with $K = W(A)$. Get analytic convex domains Ω_j with $W(A) \subset \Omega_j$ and $\sup_{\overline{\Omega}_j} \operatorname{dist}(\cdot, W(A)) \to 0$. In particular every $\overline{\Omega}_j$ lies inside the fixed bounded rectangle from Lemma **??**(ii).

Uniform bound. Fix one Ω_j. Let $M_j = \sup_{z \in \overline{\Omega}_j} \|P(z)\|$. Two cases.

If $M_j = 0$, then P vanishes on Ω_j (open), so P is the zero polynomial and $P[A] = 0$.

If $M_j > 0$, multiply P by $1/M_j$. The normalized polynomial satisfies $\|P/M_j\| \le 1$ on $\partial\Omega_j$, so Theorem **??** applies and gives $\|(P/M_j)[A]\| \le 2$, i.e., $\|P[A]\| \le 2M_j$.

In either case, $\|P[A]\| \le 2M_j$.

Shrinking. Since $\overline{\Omega}_j$ converges to $W(A)$ in Hausdorff distance and all $\overline{\Omega}_j$ lie in a fixed compact set, uniform continuity of the continuous function $z \mapsto \|P(z)\|$ implies $M_j \to \sup_{z \in W(A)} \|P(z)\| = \max_{W(A)} \|P\|$. Passing to the limit in $\|P[A]\| \le 2M_j$,

$$\|P[A]\| \;\le\; 2 \max_{z \in W(A)} \|P(z)\|.$$

This is exactly (**??**). □

Sharpness of the constant 2

Sharpness in Theorem **??**. Take $n = 2$, $m = 1$, $d = 1$, $P(z) = z$, and $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. We already saw in the Introduction that $W(A)$ is the closed unit disk, and that $\|A\| = 2$; so $\|P[A]\| = 2$ and $\max_{W(A)} |P(z)| = 1$. The ratio is

$$\frac{\|P[A]\|}{\max_{W(A)} \|P(z)\|} \;=\; \frac{2}{1} \;=\; 2.$$

Hence no smaller constant works uniformly in n, m, d. □

The constant 2 is achieved by a very asymmetric matrix: A is 2-step nilpotent, $W(A)$ is a disk (not the spectrum), and the ratio $\|A\| / \max_{W(A)} |z|$ is exactly 2. This is why the numerical range, not the spectrum, is the right set: the spectrum of A is $\{0\}$, and no bound $\|P[A]\| \le C \max_{\sigma(A)} |P|$ would hold for any finite C.

A The exterior conformal collar

This appendix supplies the classical conformal input used in Section **??**. Everything here is covered by Ahlfors [**?**]; we spell out the two ingredients that are rarely stated explicitly: real-analytic boundary continuation and global injectivity on a small exterior collar.

Lemma A.1 (Exterior coordinate and a univalent collar)**.** *Let $\Omega \subset \mathbb{C}$ be a bounded convex domain with regular real-analytic Jordan boundary. Then there are $0 < r_0 < 1$ and a map h which is meromorphic and one-to-one on $\{\lambda : |\lambda| > r_0\} \cup \{\infty\}$, has a simple pole at ∞, and maps $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$, and $|\lambda| = 1$ onto $\partial\Omega$. The expansion at infinity is*

$$h(\lambda) \;=\; c\lambda + c_0 + \sum_{j\geq 1} c_j \lambda^{-j}, \qquad c \neq 0.$$

*For λ on the unit circle, $q(\lambda) = \lambda h'(\lambda)$ is a nonzero outward normal vector at $h(\lambda)$, and the support inequality (**??**) holds.*

Proof. Step 1: A bounded simply connected model. Choose a point $z_* \in \Omega$ and apply $z \mapsto 1/(z - z_*)$ (with ∞ mapped to 0). The spherical exterior $(\mathbb{C}\setminus\overline{\Omega})\cup\{\infty\}$ maps onto a bounded simply connected domain U containing 0. Because Ω is convex, each ray from z_* meets $\partial\Omega$ in a unique point; after inversion, the exterior portions of those rays become a domain that is star-shaped about 0. Its boundary is a regular real-analytic Jordan curve: the inversion of a regular real-analytic curve is again one.

Step 2: Riemann mapping. By the Riemann mapping theorem [**?**, Ch. 6, §1.1, Thm. 1], there is a conformal bijection $g : \mathbb{D} \to U$ with $g(0) = 0$. By Carathéodory's Jordan-domain theorem [**?**, Ch. 6, §1.2], g extends to a homeomorphism $\overline{\mathbb{D}} \to \overline{U}$.

Step 3: Local real-analytic continuation across the boundary. A regular real-analytic boundary arc has a real-analytic parametrisation with nonzero derivative. Its holomorphic continuation and the holomorphic inverse function theorem give local coordinates in which the arc is a real interval and the domain is one side of that interval. Fix such coordinates at a boundary point $\zeta \in \partial\mathbb{D}$ and its image $p = g(\zeta) \in \partial U$. Continuity of the boundary extension allows shrinking the source neighborhood so that g maps its intersection with $\overline{\mathbb{D}}$ into the target chart. In these coordinates, g is holomorphic on a half-disk, continuous on the real interval, and real-valued there. The Schwarz reflection principle [**?**, Ch. 4, §6.5; Ch. 6, §§1.3–1.4] extends g holomorphically across the interval. Apply the same procedure to g^{-1}. Shrinking neighborhoods allows composing the reflected maps. Their compositions equal the identity on the original domain side, and hence throughout a smaller neighborhood by the identity theorem. So the continued g has nonzero derivative at every boundary point.

Step 4: Patching local continuations. For each $\zeta \in \partial\mathbb{D}$, choose a disk $B(\zeta, R_\zeta)$ on which such a continuation $\tilde{g}_\zeta$ exists, agreeing with g on $B(\zeta, R_\zeta) \cap \mathbb{D}$. Cover the compact circle by finitely many smaller disks $V_i = B(\zeta_i, R_i/3)$; write $U_i = B(\zeta_i, R_i)$. If $V_i \cap V_j \neq \varnothing$ and $R_i \geq R_j$, then $|\zeta_i - \zeta_j| < (R_i + R_j)/3 \leq 2R_i/3 < R_i$, so the two larger disks overlap in a connected lens $U_i \cap U_j$ that contains ζ_j and meets $\mathbb{D}$. On $\mathbb{D} \cap U_i \cap U_j$, both $\tilde{g}_i$ and $\tilde{g}_j$ agree with g, hence with each other. By the identity theorem they agree on the whole lens $U_i \cap U_j$. So the local extensions patch with g to a single holomorphic function on $\mathbb{D} \cup \bigcup_i V_i$, which is a neighborhood of the closed disk.

Step 5: Global injectivity on a small neighborhood. The extension is injective on a (possibly smaller) neighborhood of the closed disk. Otherwise there would be sequences $(w_n), (w_n')$ with $w_n \neq w_n'$, $w_n, w_n' \to$ (a neighborhood of closed disk), and $\tilde{g}(w_n) = \tilde{g}(w_n')$. Passing to subsequences inside the compact closed disk, both go to a common limit w^* in the closed disk. Since $\tilde{g}$ is continuous and injective on the closed disk (g is a homeomorphism on $\overline{\mathbb{D}}$), and $\tilde{g}(w_n) = \tilde{g}(w_n')$, we must eventually have w_n, w_n' in a common small disk about w^* on which $\tilde{g}$ is injective, contradiction. By compactness, the (injective) neighborhood is open and contains a disk $\{w : |w| < R_0\}$ for some $R_0 > 1$.

Step 6: Assembly. The function g is univalent on $\{w : |w| < R_0\} \supset \overline{\mathbb{D}}$. Its only zero there is 0,

and that zero is simple ($g'(0) \neq 0$ by conformality). Set

$$h(\lambda) \;=\; z_* + \frac{1}{g(1/\lambda)}, \qquad r_0 \;=\; R_0^{-1}.$$

Then h is meromorphic on $\{|\lambda| > r_0\} \cup \{\infty\}$, has a simple pole at ∞ (from the simple zero of g at 0), maps $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$ (from g mapping $\mathbb{D}$ to U and $z \mapsto z_* + 1/z$ undoing the inversion), and maps $|\lambda| = 1$ onto $\partial\Omega$. The expansion $h(\lambda) = c\lambda + c_0 + O(1/\lambda)$ has $c = 1/g'(0) \neq 0$.

Step 7: The support inequality. At $\lambda = e^{i\theta} \in \mathbb{T}$, the radial and angular derivatives of h are $q(\lambda) = \lambda h'(\lambda)$ and $iq(\lambda)$, respectively. They are perpendicular and nonzero. Increasing the radial parameter $|\lambda|$ traces out the exterior of $\overline{\Omega}$, so $q(\lambda)$ points outward. Since Ω is convex with a smooth boundary, the supporting-line property holds at $h(\lambda)$: for every $z \in \overline{\Omega}$,

$$\Re\Big(\overline{q(\lambda)}\big(h(\lambda) - z\big)\Big) \;\geq\; 0.$$

This is exactly (**??**). □

Sanity check for the disk

For $\Omega = \mathbb{D}$ (the unit disk), the exterior coordinate is $h(\lambda) = \lambda$. Then $q(\lambda) = \lambda$, $b_k(z) = z^k$, and $s(\lambda, \mu) = 0$. The kernel K is 1, and the marginal identities are trivial. The support inequality reads $\Re(\overline{\lambda}(\lambda - z)) = 1 - \Re(\overline{\lambda} z) \geq 0$ for $|z| \leq 1$, $|\lambda| = 1$, which is just Cauchy–Schwarz.

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